

RALPH ANDREI T. CASTILLO

Caloocan City, Metro Manila, Philippines | 0991 598 3227 | castilloralphandreit@gmail.com
LinkedIn: [linkedin.com/in/ralphcastillo02](https://www.linkedin.com/in/ralphcastillo02) | Github: github.com/andreirtc | Portfolio: <https://ralphandreicastillo.vercel.app>

PROFILE

Computer Engineering student with hands-on experience in PC repair, enterprise hardware testing, embedded-system integration, and Cisco-based network design. Led academic projects involving a standalone ATmega328P with custom PCB integration, high-availability network simulations, and VGG16 image classification. Seeking an internship where practical troubleshooting, technical documentation, and foundational hardware/networking skills can support engineering and IT operations.

TECHNICAL SKILLS

Hardware & IT Support: Windows installation and troubleshooting, desktop and laptop repair, PC assembly, component diagnostics, RAM/SSD/PSU upgrades, BIOS and driver troubleshooting, hardware testing, and system optimization

Embedded & Electronics: Arduino Uno/ATmega328P, circuit schematic design, EasyEDA PCB layout, embedded-system integration, sensor/actuator interfacing, firmware testing and troubleshooting

Networking: Cisco Packet Tracer, IPv4 addressing and subnetting, VLANs, inter-VLAN routing, OSPF, HSRP, EtherChannel, ACLs, port security

Programming & Tools: Python, Arduino C/C++, HTML/CSS/JavaScript fundamentals, Git/GitHub, Linux Mint installation and CLI basics, AutoCAD, TinkerCAD, Google Colab, Anaconda, VS Code

EXPERIENCE

Mar 2021 - Present	Computer Technician , Freelance • Provide paid, referral-based support for desktop and laptop users, resolving Windows, software, driver, performance, and hardware-related issues. • Perform Windows installation, RAM/SSD/PSU upgrades, component diagnostics, and practical repair recommendations based on client needs and budgets. • Handle advanced hardware inspection and repair tasks, including careful CPU pin assessment and alignment when feasible.
Jul 2025 - Sep 2025	Technical Engineer Intern , Micro Pacific Technologies and Systems Corporation Makati City • Tested and documented 400+ desktops, laptops, all-in-one PCs, mini PCs, and server units to support inventory, readiness, and deployment workflows. • Performed RAM and storage upgrades, BIOS validation, Windows and driver readiness checks, defect documentation, and hardware compatibility verification; prepared 30 Ethernet cables, assembled 100+ desktops, supported server RAM upgrades, and contributed to technical reporting.
Nov 2023 - Jul 2025	Academic Tutor , Independent • Delivered personalized lessons and study materials for elementary, senior high school, and college students, translating complex technical concepts into understandable steps.

PROJECTS

SolAlign: Dual-Axis Solar Tracking and Monitoring System <i>Microprocessors</i> <i>Project Lead / Embedded Hardware Lead</i>	
• Led hardware integration for a custom ATmega328P dual-axis solar tracker; independently designed the schematic and co-developed the EasyEDA PCB and firmware for LDR/servo control, manual/stow operation, LCD/keypad access, EEPROM, and INA219 monitoring.	
Enterprise Network Design and Security <i>Data & Digital Communications / Computer Networks and Security</i> <i>Project Lead</i>	
• Designed and configured a Cisco Packet Tracer 3-tier enterprise network with 33 endpoints, VLANs, OSPF, HSRP, LACP EtherChannel, ACLs, port security, WLC management, and Syslog monitoring.	
• Designed the physical network floor plan, device and rack placement, cable routes, MDF/IDF structure, Wi-Fi coverage, and redundant fiber backbone for a two-storey facility supporting 60+ endpoints.	
MRI Brain Tumor Image Classification Using VGG16 <i>Pattern Recognition</i> <i>Project Lead / Machine Learning Developer</i>	
• Developed and evaluated a VGG16 transfer-learning model in Python and Google Colab using 7,200 MRI images across four classes; achieved 89.56% test accuracy, 89.61% weighted precision, 89.56% weighted recall, and 89.31% weighted F1-score.	
Windsurf: Interactive Data Structures and Algorithms Learning Platform <i>Data Structures and Algorithms</i> <i>Project Lead / Full Stack Developer</i>	
• Led a six-member team and co-developed a web-based learning platform with a Metro Manila rail route visualizer featuring route/time display, station animation, and destination voice alerts using HTML, CSS, JavaScript, React, and Python.	
ActiveCampus Go <i>PUP Technology Festival 2025 Hackathon</i> <i>UI and Visual Asset Contributor</i>	
• Created pixel-art visual assets and contributed interface concepts for a gamified, location-based campus fitness platform during the PUP Digital Bayanihan hackathon.	

EDUCATION

Sep 2023 - Present	Bachelor of Science in Computer Engineering , Polytechnic University of the Philippines - Sta. Mesa, Manila <i>President's and Dean's Lister (multiple semesters) Expected Graduation: September 2027</i>
--------------------	---

CERTIFICATIONS

Sep 2025 - Feb 2026	CCNA 1: Introduction to Networks , Cisco Networking Academy
Feb 2026 - June 2026	CCNA 2: Switching, Routing, and Wireless Essentials , Cisco Networking Academy